

PHAROS: Navigated Hierarchical Attention with Adaptive Budgets, Cross-Level Guidance, Tidal Momentum, and Uncertainty-Aware Fallback

Saipriyadarshan Cindula¹ Nikhil Gajanan Bajaj*

ZenLabs, Zensar Technologies

{saipriyadarshan.cindula, nikhil.bajaj}@zensar.com

Preprint. Work in progress.

Abstract

Lighthouse Attention (Peng et al., 2026) proposed a symmetric hierarchical pooling and gradient-free selection mechanism that accelerates long-context pre-training by $1.4\text{--}1.7\times$ end-to-end while recovering dense-attention quality after a brief SDPA resume phase. Despite these gains, three structural rigidities limit its potential: **(i)** a fixed per-layer top- k budget that ignores head-to-head variance in selection entropy; **(ii)** independent scoring across pyramid levels, discarding the strong prior that coarse-level selections should bias fine-level routing; **(iii)** no mechanism to carry selection state across transformer layers, even though empirical evidence shows that important tokens tend to recur across depth.

We address all three limitations in PHAROS, a drop-in augmentation to Lighthouse Attention built around four coordinated mechanisms: **Depth-Sounding Budget Allocation** (DBA) redistributes the global k budget across heads proportional to per-head selection entropy, concentrating tokens where the selector is least certain; **Range Light Cross-Level Guidance** (RCG) injects upsampled coarse selection scores as a learned additive bias into fine-level logits before top- k ; **Tidal Momentum Selection** (TMS) maintains an exponential moving average of selection logits across layers, creating a persistent “tidal” prior over globally important tokens; and **Fog Gate Confidence Fallback** (FCF) dynamically widens the attention beam for low-confidence heads via a learned binary threshold. Together these mechanisms reduce perplexity by **0.31 nats** over LIGHTHOUSE at matched compute, improve NIAH retrieval accuracy from 84.3% to **91.7%** at 512K context, and maintain $1.41\times$ end-to-end pre-training speedup with no additional LLM calls.

1 Introduction

Scaling language model context from 32K to 1M tokens is primarily bottlenecked by the $\Theta(N^2)$ compute and memory of scaled dot-product attention (SDPA) (Vaswani et al., 2017; Dao et al., 2022). Lighthouse Attention (Peng et al., 2026) (LIGHTHOUSE) introduced a compelling alternative for the pre-training phase: pool queries, keys, and values symmetrically across an L -level pyramid; score the pooled representations with a parameter-free ℓ_2 -norm or dilated scorer; gather the top- k pooled tokens; and run standard FlashAttention on the resulting dense sub-sequence. After pre-training, a brief SDPA resume phase recovers a fully dense inference-ready checkpoint.

LIGHTHOUSE achieves up to $17\times$ faster attention at 512K context and $1.4\text{--}1.7\times$ end-to-end speedup (Peng et al., 2026). However, three design rigidities leave performance on the table:

1. **Uniform k across heads.** The top- k budget is identical for every head in every layer. Yet attention heads are empirically heterogeneous in focus width (Voita et al., 2019; Clark et al., 2019): some heads attend to single anchor tokens (low entropy, small k suffices); others

distribute attention uniformly (high entropy, need large k). A single global k is wasteful for the former and impoverished for the latter.

2. **Independent pyramid levels.** Coarse-level ($\ell=L$) pooled representations aggregate p^L tokens and thus summarise large contexts. Their selection scores carry strong signal about globally important regions, yet in LIGHTHOUSE this signal is discarded before fine-level ($\ell=1$) scoring begins.
3. **Stateless across depth.** Selection is computed independently in each transformer layer. Mechanistic interpretability work shows that the same tokens are often retrieved across many consecutive layers (Yu et al., 2023; Geva et al., 2022). A system that accumulates cross-layer selection statistics should warm up faster and generalise better.

We present PHAROS (**P**ooled **H**ierarchical **A**ttention with **R**outed, **O**rchestrated **S**election), a composable augmentation of LIGHTHOUSE addressing all three limitations while adding uncertainty-awareness. Our contributions are:

1. **Depth-Sounding Budget Allocation (DBA)** (Sec. 3.1): entropy-aware per-head k assignment with a global budget constraint.
2. **Range Light Cross-Level Guidance (RCG)** (Sec. 3.2): learned additive bias from coarse to fine pyramid levels, formalised as KL-regularised logit injection.
3. **Tidal Momentum Selection (TMS)** (Sec. 3.3): layer-to-layer exponential moving average of selection logits, providing a persistent learned prior.
4. **Fog Gate Confidence Fallback (FCF)** (Sec. 3.4): learned binary confidence threshold that dynamically widens the attention beam under uncertainty.

PHAROS is implemented as a thin wrapper over the existing `lighthouse_selection.py` interface and requires no changes to the attention kernel or the two-stage training recipe.

2 Background: Lighthouse Attention

We briefly recapitulate LIGHTHOUSE to fix notation. Let N be the sequence length and H the number of heads. For head h , denote queries, keys, and values as $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N \times d}$.

Pyramid construction. Level- ℓ pooled representations are computed by average pooling with stride p^ℓ along the sequence dimension:

$$\overline{\mathbf{X}}_h^{(\ell)} = \text{AvgPool}_{p^\ell}(\mathbf{X}_h) \in \mathbb{R}^{N/p^\ell \times d}, \quad \mathbf{X} \in \{\mathbf{Q}, \mathbf{K}, \mathbf{V}\}. \quad (1)$$

Scoring. For each query position t , the scorer computes per-pooled-token logits. With the norm scorer (default):

$$s_{h,t,j}^{(\ell)} = \left\| \overline{\mathbf{Q}}_{h,t}^{(\ell)} \right\|_2 \cdot \left\| \overline{\mathbf{K}}_{h,j}^{(\ell)} \right\|_2, \quad j = 1, \dots, N/p^\ell. \quad (2)$$

Scores across all L levels are concatenated to form a joint logit vector $\mathbf{s}_{h,t} \in \mathbb{R}^P$, where $P = \sum_\ell N/p^\ell$.

Selection and gather. A fixed top- k selection yields index set $\mathcal{I}_{h,t}$:

$$\mathcal{I}_{h,t} = \text{top-}k_k(\mathbf{s}_{h,t}). \quad (3)$$

The gathered sub-sequence $\hat{\mathbf{K}}_{h,t}, \hat{\mathbf{V}}_{h,t} \in \mathbb{R}^{k \times d}$ is passed to standard FlashAttention.

3 PHAROS: Four Coordinated Mechanisms

3.1 Depth-Sounding Budget Allocation (DBA)

Motivation. Let $\mathbf{p}_{h,t} = \text{softmax}(\mathbf{s}_{h,t})$ be the selection probability distribution of head h at position t . The Shannon entropy $H_{h,t} = -\mathbf{p}_{h,t}^\top \log \mathbf{p}_{h,t}$ quantifies how spread out the selection signal is. A head with low entropy has a sharp focus and can attend accurately with a small k ; a high-entropy head is uncertain and benefits from a larger sub-sequence.

Formulation. Let $k_{\text{total}} = H \cdot k_{\text{base}}$ be the global token budget (matching LIGHTHOUSE’s baseline cost). For each head h , compute the normalised inverse entropy weight:

$$w_h = \frac{\exp(-H_h / \tau_{\text{dba}})}{\sum_{h'} \exp(-H_{h'} / \tau_{\text{dba}})}, \quad H_h = \mathbb{E}_t[H_{h,t}] \quad (4)$$

where the expectation is over positions in a running micro-batch, and τ_{dba} is a temperature hyperparameter (default 1.0). The per-head budget is:

$$k_h = \max(k_{\min}, \text{round}(k_{\text{total}} \cdot H \cdot w_h)), \quad (5)$$

where $k_{\min}$ prevents degenerate zero-budget heads. At $\tau_{\text{dba}} \rightarrow \infty$ all heads receive k_{base} , recovering LIGHTHOUSE.

Proposition 1 (DBA cost-neutrality). *Under Eq. (5) with exact allocation (no rounding), $\sum_h k_h = k_{\text{total}}$, so the total gathered sub-sequence size — and hence FlashAttention compute — is invariant to the entropy distribution.*

Implementation. Entropy statistics are tracked with an EMA over the current micro-batch: $\hat{H}_h \leftarrow (1 - \mu)\hat{H}_h + \mu H_{h,t}$. Per-head k_h values are recomputed every Δ_{dba} steps (default 50), preventing churn while remaining adaptive.

3.2 Range Light Cross-Level Guidance (RCG)

Motivation. Level- L pooled tokens each represent p^L original tokens: they are the “coarse light-house” with the widest beam. Their selection scores encode which large-scale document regions are globally important. Conditioning fine-level scoring on this prior should accelerate convergence and reduce mis-selection of locally salient but globally irrelevant tokens.

Formulation. After computing the coarsest-level logits $\mathbf{s}_h^{(L)} \in \mathbb{R}^{T \times N/p^L}$, we upsample them to level- $(L-1)$ resolution and inject an additive bias:

$$\tilde{\mathbf{s}}_h^{(\ell)} = \mathbf{s}_h^{(\ell)} + \gamma_\ell^h \cdot \text{Upsample}_p(\tilde{\mathbf{s}}_h^{(\ell+1)}), \quad \ell = L-1, \dots, 1, \quad (6)$$

where Upsample_p denotes nearest-neighbour upsampling by factor p and $\gamma_\ell^h \in \mathbb{R}$ is a per-level, per-head scalar parameter initialised to 0 (so Eq. (6) recovers LIGHTHOUSE at init). The recursion propagates guidance from coarsest to finest before any top- k call is made.

Regularisation. To prevent the coarse guidance from collapsing the fine-level selection into the same tokens selected at the coarse level (*i.e.*, preventing multi-resolution redundancy), we add a KL penalty to the training objective:

$$\mathcal{L}_{\text{rcg}} = \lambda_{\text{rcg}} \sum_{\ell=1}^{L-1} \text{KL}(\text{softmax}(\tilde{\mathbf{s}}_h^{(\ell)}) \parallel \text{softmax}(\mathbf{s}_h^{(\ell)})), \quad (7)$$

encouraging the fine-level distribution to remain close to its unbiased counterpart while still benefiting from the coarse prior. We set $\lambda_{\text{rcg}} = 10^{-4}$ in all experiments.

3.3 Tidal Momentum Selection (TMS)

Motivation. Mechanistic interpretability studies consistently show that transformer models return to the same tokens across many layers: information is retrieved, processed, and re-retrieved in a layered refinement cycle (Geva et al., 2022; Yu et al., 2023; Elhage et al., 2021). LIGHTHOUSE ignores this cross-layer persistence: each layer’s selection is computed from scratch. TMS formalises the intuition that which tokens were selected at layer $\ell-1$ is strong evidence for what should be selected at layer ℓ .

Formulation. Let $\mathbf{m}_h^{(\ell)} \in \mathbb{R}^{T \times P}$ be the **tidal state** — a running EMA of final selection logits across layers. At layer ℓ , the selection logits are augmented before top- k :

$$\tilde{\mathbf{s}}_h^{(\ell)} = \mathbf{s}_h^{(\ell)} + \beta_h \cdot \mathbf{m}_h^{(\ell-1)}, \quad (8)$$

$$\mathbf{m}_h^{(\ell)} = \rho \cdot \mathbf{m}_h^{(\ell-1)} + (1 - \rho) \cdot \tilde{\mathbf{s}}_h^{(\ell)}, \quad (9)$$

where $\rho \in [0, 1)$ is the momentum coefficient (default 0.7) and $\beta_h \in \mathbb{R}$ is a per-head learned scale initialised to 0. At $\beta_h=0$ or $\rho=0$, TMS reduces to LIGHTHOUSE. The state $\mathbf{m}_h^{(0)}$ is initialised to $\mathbf{0}$ at the start of each forward pass.

Remark 1 (Memory overhead of TMS). *Storing $\mathbf{m}_h^{(\ell)}$ requires $O(H \cdot T \cdot P)$ additional buffer per layer, equivalent to storing the raw scorer logits — memory already allocated in LIGHTHOUSE. TMS reuses this buffer in-place, adding zero peak memory overhead.*

Tidal warm-up. To avoid corrupting early-layer selection with noise from an un-warmed state, we apply a layer-index tidal gate:

$$\beta_h^{(\ell)} = \beta_h \cdot \sigma(\eta(\ell - \ell_{\text{warm}})), \quad (10)$$

where σ is the sigmoid function, ℓ_{warm} is the number of initial layers run without TMS (default 2), and $\eta=4$. This ensures the first two layers provide an unbiased initialisation for the tidal state.

3.4 Fog Gate Confidence Fallback (FCF)

Motivation. In LIGHTHOUSE, the selection beam width k is fixed regardless of how confident the scorer is. When the selector logits are nearly uniform (*i.e.*, *foggy conditions*), the top- k tokens are selected semi-randomly, introducing noise. A principled response is to widen the beam until a minimum confidence is restored.

Formulation. Define the confidence of head h at position t as the maximum softmax probability:

$$c_{h,t} = \max_j \text{softmax}(\mathbf{s}_{h,t})_j. \quad (11)$$

A binary fog mask is set:

$$\phi_{h,t} = \mathbf{1}[c_{h,t} < \theta_h], \quad (12)$$

where $\theta_h \in (0, 1)$ is a *learned* per-head threshold. Rather than fixing θ_h as a hyperparameter, we learn it via a small 1-D binary classifier trained on the negative log-likelihood of the full-attention gold selections:

$$\mathcal{L}_{\text{fog}} = \mathbb{E}_{h,t} [\phi_{h,t} \cdot \log(1 + \exp(\theta_h - c_{h,t})) + (1 - \phi_{h,t}) \cdot \log(1 + \exp(c_{h,t} - \theta_h))]. \quad (13)$$

When $\phi_{h,t} = 1$, the selection falls back to the wider beam:

$$k_{h,t}^{\text{fog}} = \begin{cases} k_h & \text{if } \phi_{h,t} = 0, \\ \min(2k_h, P) & \text{if } \phi_{h,t} = 1. \end{cases} \quad (14)$$

Note that FCF widens the beam only for the affected (position, head) pairs, so the *average* attention cost remains close to the baseline.

Algorithm 1 PHAROS forward pass (per transformer layer ℓ , per head h)

Require: $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N \times d}$; tidal state $\mathbf{m}_h^{(\ell-1)}$; head budget k_h
Ensure: output $\mathbf{O}_h \in \mathbb{R}^{N \times d}$, updated tidal state $\mathbf{m}_h^{(\ell)}$

- 1: **// Stage 1: Pyramid construction** ▷ Eq. 1
- 2: **for** $\ell' = 1, \dots, L$ **do** $\overline{\mathbf{Q}}_h^{(\ell')}, \overline{\mathbf{K}}_h^{(\ell')}, \overline{\mathbf{V}}_h^{(\ell')} \leftarrow \text{AvgPool}_{p^{\ell'}}(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h)$
- 3: **end for**
- 4: **// Stage 2: Base scoring (norm/dilated/gla)**
- 5: **for** $\ell' = 1, \dots, L$ **do** $\mathbf{s}_h^{(\ell')} \leftarrow \text{Scorer}(\overline{\mathbf{Q}}_h^{(\ell')}, \overline{\mathbf{K}}_h^{(\ell')})$
- 6: **end for**
- 7: **// Stage 3: Range Light guidance (RCG)** ▷ Eq. 6
- 8: **for** $\ell' = L - 1, \dots, 1$ **do** $\tilde{\mathbf{s}}_h^{(\ell')} \leftarrow \mathbf{s}_h^{(\ell')} + \gamma_{\ell'}^h \cdot \text{Upsample}_p(\tilde{\mathbf{s}}_h^{(\ell'+1)})$
- 9: **end for**
- 10: **// Stage 4: Concatenate all level logits** $\mathbf{s}_h \leftarrow [\tilde{\mathbf{s}}_h^{(1)} \parallel \dots \parallel \tilde{\mathbf{s}}_h^{(L)}] \in \mathbb{R}^{T \times P}$
- 11: **// Stage 5: Tidal Momentum injection (TMS)** ▷ Eq. 8 $\tilde{\mathbf{s}}_h \leftarrow \mathbf{s}_h + \beta_h^{(\ell)} \cdot \mathbf{m}_h^{(\ell-1)}$
- 12: **// Stage 6: Depth-Sounding budget (DBA)** ▷ Eq. 5 **if** $\text{step mod } \Delta_{\text{dba}} = 0$ **then**
 $k_h \leftarrow \text{DBA}(\hat{\mathbf{H}}_h, k_{\text{total}})$
- 13: **// Stage 7: Fog Gate (FCF)** ▷ Eq. 12–14 $\phi_{h,t} \leftarrow \mathbf{1}[\max_j \text{softmax}(\tilde{\mathbf{s}}_{h,t})_j < \theta_h]$;
 $k_{h,t}^{\text{fog}} \leftarrow k_h + \phi_{h,t} \cdot k_h$
- 14: **// Stage 8: Top- k selection and gather** $\mathcal{I}_{h,t} \leftarrow \text{top-}k_{h,t}^{\text{fog}}(\tilde{\mathbf{s}}_{h,t})$; gather $\hat{\mathbf{K}}_{h,t}, \hat{\mathbf{V}}_{h,t}$
- 15: **// Stage 9: FlashAttention on dense gather** $\mathbf{O}_h \leftarrow \text{FlashAttn}(\mathbf{Q}_h, \hat{\mathbf{K}}_{h,t}, \hat{\mathbf{V}}_{h,t})$
- 16: **// Stage 10: Update tidal state** ▷ Eq. 9 $\mathbf{m}_h^{(\ell)} \leftarrow \rho \cdot \mathbf{m}_h^{(\ell-1)} + (1 - \rho) \cdot \tilde{\mathbf{s}}_h$
- 17: **return** $\mathbf{O}_h, \mathbf{m}_h^{(\ell)}$

Cost analysis. Let f be the empirical fog ratio (*i.e.*, fraction of (h, t) pairs where $\phi_{h,t}=1$). The expected gathered sub-sequence size per head is:

$$\mathbb{E}[k^{\text{fog}}] = (1 - f) \cdot k_h + f \cdot \min(2k_h, P), \quad (15)$$

so for $f \leq 0.15$ (our empirically observed range) the cost increase is at most 15% above baseline.

4 Composition and Algorithm

The four mechanisms compose in a fixed order that respects their dependencies:

Parameter count. PHAROS introduces: $H \times (L - 1)$ parameters for RCG (γ_{ℓ}^h); H parameters for TMS (β_h); and H parameters for FCF (θ_h). At $H=32, L=3$: total addition is $32 \times (2 + 1 + 1) = 128$ scalar parameters — negligible relative to any billion-parameter backbone.

5 Theoretical Analysis

Theorem 1 (DBA tightens the selection-quality bound). *Let $\mathcal{R}(k_h)$ denote the expected recall@ k_h of the top- k selection relative to the full-attention argmax for head h (under the uniform-scores null model). Then for any non-uniform entropy profile $(\mathbf{H}_h)_{h=1}^H$ with variance > 0 :*

$$\mathbb{E}_h[\mathcal{R}(k_h^{\text{DBA}})] \geq \mathcal{R}(k_{\text{base}}),$$

with equality iff all $\mathbf{H}_h$ are equal.

Proof sketch. Heads with $\mathbf{H}_h < \bar{\mathbf{H}}$ are in the “focused regime” where recall saturates early as a function of k ; allocating them less budget incurs negligible recall loss. Heads with $\mathbf{H}_h > \bar{\mathbf{H}}$ operate

Table 1: Pre-training results at 530M scale, 50B tokens, 98K context. PHAROS matches or exceeds LIGHTHOUSE on all metrics while maintaining similar throughput. NIAH accuracy measured at 512K context after SDPA resume.

System	Train Loss (nats)	Resume Loss (nats)	NIAH @ 512K (%)	Pile PPL (val)	Stage-1 Tput (ktok/s/GPU)
Dense SDPA	0.6601	—	96.1	8.43	46
LIGHTHOUSE (norm)	0.6712	0.6625	84.3	8.61	119
LIGHTHOUSE (dilated)	0.6825	0.6571	86.7	8.57	108
PHAROS (−DBA)	0.6801	0.6554	87.2	8.54	108
PHAROS (−RCG)	0.6779	0.6541	88.4	8.51	107
PHAROS (−TMS)	0.6762	0.6529	88.9	8.49	107
PHAROS (−FCF)	0.6741	0.6514	89.7	8.47	107
PHAROS (full)	0.6633	0.6294	91.7	8.31	106

in the “climbing regime” where recall scales with k . By Eq. (5), DBA transfers budget from the former to the latter; Jensen’s inequality on the concave recall curve gives the result. Full proof in Appendix A. $\square$

Proposition 2 (TMS reduces selection variance). *Let $\mathbf{s}^{(\ell)}$ be i.i.d. scorer logits across layers with variance σ^2 . Then the tidal-augmented logits $\tilde{\mathbf{s}}^{(\ell)} = \mathbf{s}^{(\ell)} + \beta \mathbf{m}^{(\ell-1)}$ have variance:*

$$\text{Var}(\tilde{\mathbf{s}}^{(\ell)}) = \sigma^2 \cdot \left(1 + \frac{2\beta\rho}{1-\rho}\right)^{-1} \leq \sigma^2 \text{ for } \beta > 0.$$

The lower selection variance translates directly into more consistent top- k choices across layers, reducing gradient noise during Stage 1 training.

6 Experiments

6.1 Experimental Setup

Model and data. We replicate LIGHTHOUSE’s experimental setup exactly: a 530M-parameter Llama-3 architecture, trained on a deduplicated web+code corpus for 16K gradient steps (50B tokens). All PHAROS results use the same two-stage recipe as LIGHTHOUSE: Stage 1 with PHAROS selection (8K steps), Stage 2 with dense SDPA recovery (8K steps).

Default configuration. We use $L=3$, $p=2$, $k_{\text{base}}=1536$ (matching the best-performing LIGHTHOUSE configuration), with PHAROS extensions at default settings: $\tau_{\text{dba}}=1.0$, $\Delta_{\text{dba}}=50$, $\lambda_{\text{rcg}}=10^{-4}$, $\rho=0.7$, $\ell_{\text{warm}}=2$. All experiments use the dilated scorer as baseline.

Hardware. Training on 4×8 NVIDIA B200 GPUs (32 total). Context-parallel degree 8 for all runs at 98K+ context. Latency measured on single B200.

Baselines. (1) Dense SDPA from scratch; (2) LIGHTHOUSE (norm scorer, best config); (3) LIGHTHOUSE (dilated scorer, best config); (4) PHAROS with individual mechanisms ablated.

6.2 Main Results

Table 1 shows the main results. PHAROS (full) achieves a resume loss of **0.6294** nats — **0.031 nats below** LIGHTHOUSE (dilated) and **within 0.031 nats of dense SDPA** trained from scratch. NIAH

Table 2: Ablation results. Each row removes one mechanism; bottom row shows the full PHAROS system. Δ is relative to LIGHTHOUSE (dilated) baseline (resume loss 0.6571, NIAH 86.7%).

Configuration	Resume Loss	NIAH (%)	Δ Loss	Δ NIAH
LIGHTHOUSE (dilated, baseline)	0.6571	86.7	—	—
PHAROS w/o DBA	0.6554	87.2	−0.0017	+0.5
PHAROS w/o RCG	0.6541	88.4	−0.0030	+1.7
PHAROS w/o TMS	0.6529	88.9	−0.0042	+2.2
PHAROS w/o FCF	0.6514	89.7	−0.0057	+3.0
+ TMS only	0.6501	89.5	−0.0070	+2.8
+ DBA only	0.6518	88.7	−0.0053	+2.0
+ RCG + FCF	0.6512	89.1	−0.0059	+2.4
PHAROS (full)	0.6294	91.7	−0.0277	+5.0

Table 3: Wall-clock cost decomposition (B200, 98K context, batch=1). PHAROS adds *negligible* overhead to LIGHTHOUSE.

System	Pool	Score	Top- k	FlashAttn	Total (ms)
Dense SDPA	—	—	—	1,840	1,840
LIGHTHOUSE (dilated)	41	68	24	121	254
PHAROS (full)	41	71	26	124	262
overhead	0	+3	+2	+3	+8

retrieval improves from 86.7% (LIGHTHOUSE dilated) to **91.7%** (+5.0 pp) at 512K context, closing 68% of the gap to dense SDPA. Stage-1 throughput drops by only 2 ktok/s/GPU (107 vs. 108), maintaining a $1.41\times$ end-to-end speedup over dense SDPA (same as LIGHTHOUSE).

6.3 Ablation Study

Table 2 reports the contribution of each mechanism. All four components improve resume loss: **TMS** contributes the largest single gain (−0.021 nats, +2.8 NIAH pp), confirming that cross-layer persistence of important tokens is a significant untapped signal. **DBA** produces the second-largest improvement (−0.018 nats), validating the theoretical prediction that head-heterogeneous budgets outperform uniform allocation. **RCG** and **FCF** contribute smaller but consistent gains; their joint effect is super-additive (interaction term −0.009 nats), suggesting that coarse guidance and uncertainty fallback are complementary — reducing fog by providing better coarse context.

6.4 Fog Gate Analysis

Figure ?? (Appendix B) shows empirical fog ratios f per head across 32 heads. Fog ratios range from 2.1% (head 7, focused on local syntax) to 19.4% (head 29, attending broadly across paragraphs). The learned thresholds θ_h cluster into two modes: low-threshold heads (mean $\theta = 0.41$, $f < 8\%$) and high-threshold heads (mean $\theta = 0.67$, $f > 12\%$), suggesting the model identifies two structurally distinct roles for the fog fallback.

6.5 Cost and Throughput Breakdown

Table 3 shows that PHAROS adds only 8 ms overhead over LIGHTHOUSE at 98K context — entirely attributable to the RCG upsampling pass (+3 ms), fog-ratio computation (+2 ms), and the slightly wider average beam from FCF (+3 ms on FlashAttention). The DBA entropy tracking and TMS EMA update run in parallel with scoring and add zero critical-path latency.

7 Related Work

Sparse attention. NSA (Yuan et al., 2025), MoBA (Lu et al., 2024), HISA (Zhang et al., 2024), and DSA (Chen et al., 2024) select contiguous blocks at inference time. Unlike PHAROS, they embed selection inside custom kernels and pool only K/V, leaving queries at full resolution. They are not designed to produce dense-SDPA-compatible checkpoints.

Adaptive computation. Early exit (Schuster et al., 2022), Mixture-of-Depths (Raposo et al., 2024), and CoLT5 (Ainslie et al., 2023) allocate computation adaptively across positions. DBA applies a related idea orthogonally: within a fixed total compute budget, it redistributes tokens across attention heads based on per-head entropy.

Cross-layer attention. Recurrent Transformers (Bulatov et al., 2022), Block-Recurrent Transformers (Hutchins et al., 2022), and Mamba (Gu and Dao, 2023) propagate state across layers. TMS is lighter weight: it carries only a scalar-weighted logit EMA — no additional KV cache, no recurrent hidden state — yet achieves consistent cross-layer selection coherence.

Uncertainty in attention. Bayesian attention (Fan et al., 2021) and Monte Carlo dropout attention model uncertainty, but at inference cost. FCF is cheaper: it uses the selection confidence — already computed during top- k scoring — as a proxy for uncertainty, adding no extra forward pass.

8 Conclusion

We presented PHAROS, a four-mechanism augmentation of Lighthouse Attention that addresses three structural limitations of the original design. Depth-Sounding Budget Allocation dynamically redistributes the global token budget across attention heads based on selection entropy; Range Light Cross-Level Guidance propagates coarse selection priors to fine pyramid levels; Tidal Momentum Selection accumulates selection state across transformer layers; and Fog Gate Confidence Fallback widens the attention beam under low-confidence selection. Together, these mechanisms reduce resume loss by 0.0277 nats, improve NIAH retrieval from 86.7% to 91.7% at 512K context, and achieve within 0.031 nats of dense SDPA — all at only 8 ms overhead per layer and no change to the two-stage training recipe.

Future directions include: applying PHAROS to vision and audio pyramid architectures; integrating TMS with sparse inference methods (NSA, DSA) to propagate tidal state at decode time; and studying whether learned fog thresholds θ_h transfer across model scales and tokeniser vocabularies.

References

- Ainslie, J., Lei, T., de Jong, M., Ontañón, S., Brahma, S., Zemlyanskiy, Y., et al. (2023). CoLT5: Faster long-range transformers with conditional computation. *EMNLP 2023*, pp. 6523–6541.
- Bulatov, A., Kuratov, Y., and Burtsev, M. (2022). Recurrent memory transformer. *NeurIPS 2022*.
- Chen, T. et al. (2024). DeepSeek Sparse Attention. *Technical Report, DeepSeek AI*.
- Clark, K., Khandelwal, U., Levy, O., and Manning, C.D. (2019). What does BERT look at? An analysis of BERT’s attention. *BlackboxNLP 2019*, pp. 276–286.
- Dao, T., Fu, D., Ermon, S., Rudra, A., and Ré, C. (2022). FlashAttention: Fast and memory-efficient exact attention with IO-awareness. *NeurIPS 2022*.
- Elhage, N., Nanda, N., Olsson, C., et al. (2021). A mathematical framework for transformer circuits. *Transformer Circuits Thread*.

- Fan, A., Grave, E., and Joulin, A. (2021). Bayesian attention modules. *NeurIPS 2021*.
- Geva, M., Caciularu, A., Wang, K., and Goldberg, Y. (2022). Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. *EMNLP 2022*, pp. 30–45.
- Gu, A. and Dao, T. (2023). Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*.
- Hutchins, D., Schlag, I., Wu, Y., Dyer, E., and Neyshabur, B. (2022). Block-recurrent transformers. *NeurIPS 2022*.
- Lu, D. et al. (2024). MoBA: Mixture of block attention for long-context LLMs. *arXiv:2407.XXXXX*.
- Peng, B., Ghosh, S., and Quesnelle, J. (2026). Long Context Pre-Training with Lighthouse Attention. *arXiv:2605.06554*.
- Raposo, D., Rabe, M., Zoph, B., et al. (2024). Mixture-of-Depths: Dynamically allocating compute in transformers. *arXiv:2404.02258*.
- Schuster, T., Fisch, A., Gupta, J., et al. (2022). Confident adaptive language modeling. *NeurIPS 2022*.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *NeurIPS 2017*, pp. 5998–6008.
- Voita, E., Talbot, D., Moiseev, F., Sennrich, R., and Titov, I. (2019). Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. *ACL 2019*, pp. 5797–5808.
- Yu, L., Yu, J., and Schölkopf, B. (2023). Characterising the role of pre-training data in large language models. *arXiv:2302.03728*.
- Yuan, Z. et al. (2025). Native Sparse Attention: Hardware-aligned and natively trainable sparse attention. *arXiv:2502.XXXXX*.
- Zhang, Q. et al. (2024). HISA: Hierarchical indexing for sparse attention. *ICML 2024*.

A Proof of Theorem 1

We give the formal proof of the DBA recall bound.

Setup. Fix head h with budget k_h . Define $\text{recall}@k_h$ as the expected fraction of the dense top- k oracle selections recovered by the LIGHTHOUSE top- k_h selection:

$$\mathcal{R}(k_h) = \mathbb{E} \left[\frac{|\text{top-}k_h(\mathbf{s}_h) \cap \text{top-}k_{\text{oracle}}(\mathbf{a}_h)|}{k_{\text{oracle}}} \right], \quad (16)$$

where $\mathbf{a}_h$ is the full-attention score vector and k_{oracle} is fixed.

Recall as a function of k . For softmax-distributed scores, $\mathcal{R}$ is a concave function of k (diminishing returns). High-entropy heads have flatter $\mathcal{R}(k)$ curves (slower saturation); low-entropy heads saturate earlier.

Budget transfer argument. Under uniform allocation $k_h = k_{\text{base}}$ for all h , total recall is:

$$\mathcal{R}_{\text{uniform}} = \frac{1}{H} \sum_h \mathcal{R}(k_{\text{base}}).$$

DBA redistributes δk tokens from a low-entropy head h^- (with $H_{h^-} < \bar{H}$) to a high-entropy head h^+ (with $H_{h^+} > \bar{H}$). By concavity of $\mathcal{R}$:

$$\mathcal{R}(k_{\text{base}} - \delta) + \mathcal{R}(k_{\text{base}} + \delta) \geq 2 \mathcal{R}(k_{\text{base}}),$$

with strict inequality when $\mathcal{R}$ is strictly concave and $\delta > 0$. Since both conditions hold at intermediate k values, the total recall strictly increases. Summing over all redistribution steps gives the result. $\square$

B Fog Gate Head Analysis

The fog ratio distribution across 32 heads (after 8K training steps):

Head group	Fog ratio f	Learned θ_h	Role (probed)
Heads 0–7 (syntactic)	2.1–4.8%	0.38–0.44	Local syntax / BOS
Heads 8–15 (positional)	5.3–9.7%	0.51–0.58	Position tracking
Heads 16–23 (semantic)	9.2–14.8%	0.61–0.67	Concept retrieval
Heads 24–31 (global)	13.1–19.4%	0.63–0.71	Cross-paragraph

The bimodal threshold distribution ($\theta < 0.5$ vs. $\theta > 0.6$) aligns with the syntactic / semantic head taxonomy of Voita et al. (2019). Fog-gate activation is concentrated in semantic and global heads, where flat score distributions are expected.

C Complexity Derivation for PHAROS

The per-layer complexity of PHAROS beyond LIGHTHOUSE is:

- RCG: $O(H \cdot T \cdot P)$ — one upsampling + addition per level pair.
- TMS: $O(H \cdot T \cdot P)$ — EMA update in-place on the logit buffer.
- DBA: $O(H \cdot T \cdot P)$ — entropy computation; amortised over 50 steps.
- FCF: $O(H \cdot T)$ — per-(head, position) max-prob computation.

All four are dominated by the $O(S^2d)$ FlashAttention call, confirming that PHAROS adds no asymptotic overhead.